

Scalability & Future Plan

Date	28 June 2026
Team ID	SL-2026-AIML
Project Name	Smart Lender — Applicant Credibility Prediction for Loan Approval
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	No persistent database — data is read from a static CSV file	Cannot track applicant history over time	High
2	Runs on Flask's local development server	Not suitable for concurrent multi-user production traffic	Medium
3	No authentication or role separation between applicant and officer views	Anyone with the URL can submit or view predictions	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user local testing	Deploy on IBM Cloud / Render with a production WSGI server (e.g. Gunicorn)
2	Data Storage	Flat CSV file	Migrate to a relational database using the already-designed ER schema
3	Performance	Sub-second predictions locally, untested under load	Add load testing (e.g., Locust) before production rollout
4	Security	No authentication	Add officer login and role-based access before handling real applicant data

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Migrate to a relational database using the designed ER schema	Next 1-2 months	Enables applicant history and audit trail
Phase 3	Deploy to IBM Cloud with a production WSGI server	Next 2-3 months	Supports multiple concurrent users reliably
Phase 4	Add explainability (e.g. SHAP values) to show why a decision was made	Next 3-6 months	Increases officer trust and regulatory transparency